ICSE Class 10 Robotics & Artificial Intelligence

Solved 2026 Question Paper

KnowledgeBoat Solutions

2026

How to use this presentation

Each question appears on its own slide.

The next slide shows the answer and brief reason.

Images are included where the original paper had figures.

Question 1(i)

Study the image given below.

In which field is this robot used?

(a) Defence
(b) Agriculture
(c) Medicine
(d) Entertainment



Answer 1(i)

Answer

Agriculture

Reason

The figure shows a robot used in agricultural fields. Such robots perform tasks like planting seeds, harvesting crops, spraying fertilizers or pesticides, and monitoring crop health.

Question 1(ii)

.....is an example of deterministic computing.

(a) Traffic light sequencing

(b) Autonomous driving

(c) Stock market prediction

(d) Face recognition

Answer 1(ii)

Answer

Traffic light sequencing

Reason

Signals change in a fixed order based on predefined time intervals. For a given input the output is always the same and predictable.

Question 1(iii)

A room heater is an example of a/an:

(a) gear

(b) robot

(c) actuator

(d) machine

Answer 1(iii)

Answer

machine

Reason

A room heater uses electrical energy to do useful work by producing heat energy.

Question 1(iv)

Tinkercad is commonly used for:

(a) simulating robotic movements.

(b) data encryption.

(c) performing gear ratio calculations only.

(d) building operating systems.

Answer 1(iv)

Answer

simulating robotic movements.

Reason

Tinkercad is a simulation tool used to design and visualize circuits and robotic systems.

Question 1(v)

In the block diagram of a robotic control system, the correct sequence is:

(a) Input $\rightarrow$ Robot $\rightarrow$ Controller $\rightarrow$ Feedback

(b) Feedback $\rightarrow$ Input $\rightarrow$ Controller $\rightarrow$ Robot

(c) Input $\rightarrow$ Controller $\rightarrow$ Robot $\rightarrow$ Feedback

(d) Controller $\rightarrow$ Input $\rightarrow$ Robot $\rightarrow$ Feedback

Answer 1(v)

Answer

Input $\rightarrow$ Controller $\rightarrow$ Robot $\rightarrow$ Feedback

Reason

Input is given to the controller, which sends commands to the robot. Feedback monitors and adjusts the system.

Question 1(vi)

Which of the following is an example of a linear actuator?

(a) DC motor

(b) Pneumatic piston

(c) Arduino

(d) Electric fan

Answer 1(vi)

Answer

Pneumatic piston

Reason

A linear actuator produces motion in a straight line. A pneumatic piston moves back and forth using compressed air.

Question 1(vii)

Which function of Python checks whether all characters in a string 's' are alphanumeric?

(a) `s.allnum()`

(b) `s.isnumal()`

(c) `s.isalnum()`

(d) `s.isalphanum()`

Answer 1(vii)

Answer

`s.isalnum()`

Reason

`isalnum()` checks if all characters are letters or digits.

Question 1(viii)

A motor drives a small gear with 10 teeth connected to a larger gear with 40 teeth. What is the gear ratio?

(a) 4 : 1

(b) 1 : 4

(c) 1 : 2

(d) 2 : 1

Answer 1(viii)

Answer

4 : 1

Reason

Gear Ratio = teeth on driven / teeth on driving = $40/10 = 4$.

Question 1(ix)

Which string operator in Python finds the length of string 'a'?

(a) `len(a)`

(b) `len()`

(c) `a.len()`

(d) `a.len(a)`

Answer 1(ix)

Answer

```
len(a)
```

Question 1(x)

Unauthorised hardware access leads to:

(a) loss of data.

(b) increased computer speed.

(c) better performance.

(d) having more data.

Answer 1(x)

Answer

loss of data.

Question 1(xi)

In machine learning, testing data is primarily used to:

(a) train the model for better performance.

(b) measure model performance on unseen data.

(c) store training parameters.

(d) provide labels for training.

Answer 1(xi)

Answer

measure model performance on unseen data.

Question 1(xii)

The function of a sensor is to:

(a) detect changes in environment and send signals.

(b) process the input data and make decisions.

(c) change electrical energy into motion.

(d) supply power to the robot.

Answer 1(xii)

Answer

detect changes in environment and send signals.

Question 1(xiii)

The purpose of Turing Test is to evaluate:

(a) memory consumption of a computer.

(b) the processing speed of computers and humans.

(c) two computer programs.

(d) if a machine can exhibit human-like responses.

Answer 1(xiii)

Answer

if a machine can exhibit human-like responses.

Question 1(xiv)

.....software is used to simulate a robotic system.

(a) LibreOffice

(b) MS Word

(c) Tinkercad

(d) MySQL

Answer 1(xiv)

Answer

Tinkercad

Question 1(xv)

Assertion (A): Robots use actuators for movement.
Reason (R): Actuators sense the environment and convert it to electrical signals.

(a) (A) true, (R) false

(b) (A) false, (R) true

(c) Both true, R explains A

(d) Both true, R does not explain A

Answer 1(xv)

Answer

(A) is true but (R) is false.

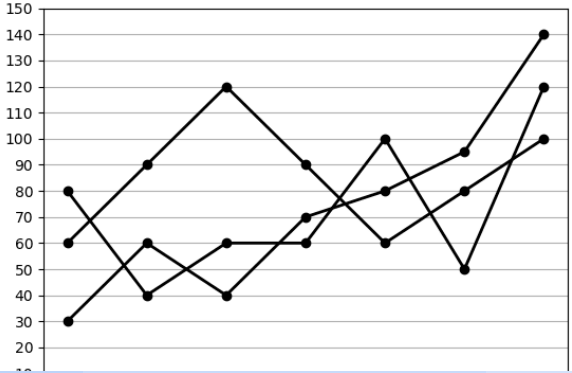
Reason

Actuators produce movement. Sensing is done by sensors.

Question 1(xvi)

Identify the type of graph shown below:

(a) Scatter plot
(b) Bar graph
(c) Pie chart
(d) Line graph



Answer 1(xvi)

Answer

Line graph

Reason

Data points are connected by continuous lines showing changes over time.

Question 1(xvii)

In AI problem scoping, "Who" refers to the:

(a) dataset used for training.

(b) end users affected by the project.

(c) programming environment used.

(d) hardware requirements of the project.

Answer 1(xvii)

Answer

end users affected by the project.

Question 1(xviii)

Give the output of the following Python code:

```
msg = "Artificial Intelligence"
print(msg[11:14])
```

Options: tel / Int / ell / gen

Answer 1(xviii)

Answer

Int

Reason

Indices 11, 12, 13 = l, n, t

Question 1(xix)

The keyword used to include a package or module in a Python program is:

(a) export

(b) include

(c) collect

(d) import

Answer 1(xix)

Answer

import

Question 1(xx)

What will be the output of the Python code given below?

```
x = "BOARD EXAMINATION"
    print(x[0:4:2])
```

Options: BOAR / BA / OR / Error

Answer 1(xx)

Answer

BA

Reason

Indices 0 and 2 = B and A

Question 2(i)

Mention any two sensors used in autonomous drones.

Answer 2(i)

Answer

GPS sensor, Ultrasonic sensor

Question 2(ii)

'In a hospital, cobots assist nurses in delivering medicines to patients.'
Give any two advantages of using cobots in hospitals.

Answer 2(ii)

Answer

Reduce the workload of nurses by assisting in routine tasks

Improve efficiency and accuracy, ensuring timely delivery

Question 2(iii)

What is spur gear? Mention one application of spur gear in robotics.

Answer 2(iii)

Answer

A spur gear has straight teeth parallel to the shaft axis and transmits motion between parallel shafts.
Application: robotic arms to transmit motion and control joint movement.

Question 2(iv)

Explain the role of data in Machine Learning.

Answer 2(iv)

Answer

Data refers to raw facts used to train models to identify patterns and relationships. It is processed into meaningful information for decisions, predictions and classifications. Quality and quantity of data directly affect model accuracy.

Question 2(v)

How is machine intelligence similar to human intelligence?

Answer 2(v)

Answer

Both improve through learning from experience. Machines learn from data; humans learn from past experiences. Both recognise patterns to analyse situations, solve problems and make decisions.

Question 2(vi)

Mention any one benefit of using Kaggle for data acquisition.

Answer 2(vi)

Answer

Kaggle provides access to a large collection of real-world datasets that can be used for training and testing machine learning models.

Question 2(vii)

Why is modelling important in AI project framework?

Answer 2(vii)

Answer

Modelling involves selecting and training an appropriate AI model using the given data to achieve the desired goal. It helps the system learn patterns and make predictions or decisions.

Question 2(viii)

What will be the output of the following Python code?

```
a = 5
b = 3
print(a > b and b < 3)
```

Answer 2(viii)

Answer

False

Reason

a $>$ b is True, b $<$ 3 is False. True AND False = False.

Question 2(ix)

What will be the output of the following Python code?

```
str1 = 'PYTHON'
print(str1[1:4])
```

Answer 2(ix)

Answer

YTH

Reason

Slice from index 1 to 3: Y, T, H.

Question 2(x)

What will be the output of the following Python code?

```
s = "Great Day"
print(s.count('a'))
print(s.startswith('g'))
```

Answer 2(x)

Answer

2
False

Reason

'a' appears twice. The string starts with 'G' (uppercase), so startswith('g') is False.

Question 3(i)

What are Assistant Robots?

Answer 3(i)

Answer

Assistant robots are intelligent machines designed to support humans in tasks such as customer service, personal assistance, healthcare and industrial work. Examples: Pepper, Roomba, CloiHomeBot, Astro.

Question 3(ii)

Give any two differences between deterministic computing and probabilistic computing with an example.

Answer 3(ii)

Answer

Deterministic: fixed and predictable output, no randomness (e.g. Calculator).
Probabilistic: output based on probability and may vary (e.g. Face recognition).

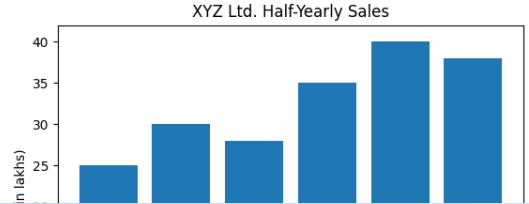
Question 3(iii)

A company XYZ Ltd. records its monthly sales in lakhs as follows:
Month = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
Sales = [25, 30, 28, 35, 40, 38]
Write a Python program using Matplotlib to store the data, plot a bar chart (month on X-axis, sales on Y-axis), label the axes and give title "XYZ Ltd. Half-Yearly Sales".

Answer 3(iii)

Answer (code)

```
import matplotlib.pyplot as plt
month = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
sales = [25, 30, 28, 35, 40, 38]
plt.bar(month, sales)
plt.xlabel("Month")
plt.ylabel("Sales (in lakhs)")
plt.title("XYZ Ltd. Half-Yearly Sales")
plt.show()
```



Question 4(i)

What are smart homes? Name any two features that make a home a smart home.

Answer 4(i)

Answer

Smart homes are homes equipped with smart devices connected through the internet and controllable remotely.
Features: Automated lighting and appliances; Smart security systems with cameras and sensors.

Question 4(ii)

Briefly explain Data Exploration from AI project framework.

Answer 4(ii)

Answer

Data Exploration is the stage where collected data is examined to understand structure, patterns, trends and relationships. It helps identify insights, detect anomalies, and select a suitable model using visualisation tools.

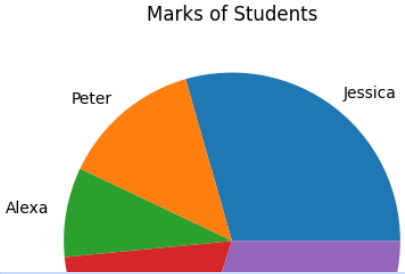
Question 4(iii)

Write a Python program to plot a pie chart comparing marks of five students:
stud_name = ['Jessica', 'Peter', 'Alexa', 'James', 'Ram']
marks = [48, 22, 14, 30, 49]

Answer 4(iii)

Answer (code)

```
import matplotlib.pyplot as plt
stud_name = ['Jessica', 'Peter', 'Alexa', 'James', 'Ram']
marks = [48, 22, 14, 30, 49]
plt.pie(marks, labels=stud_name)
plt.title("Marks of Students")
plt.show()
```



Question 5(i)

Name the type of sensor used for each of the following purposes:
(a) Detecting obstacles
(b) Measuring light intensity
(c) Measuring temperature

Answer 5(i)

Answer

(a) Ultrasonic sensor
(b) Light sensor
(c) Temperature sensor

Question 5(ii)

Explain software theft. Mention any two consequences of software theft.

Answer 5(ii)

Answer

Software theft is the unauthorised use of software by copying or modifying it (illegal downloading, piracy).
Consequences: financial losses for developers; legal offence leading to punishments.

Question 5(iii)

Write a Python program to:
Input a sentence and store it in a string.
Print the length, first character and last character.
Use a loop to count and print the total number of digits present.

Answer 5(iii)

Answer (code outline)

Input sentence, print length / first / last character, loop with `isdigit()` to count digits.

Question 6(i)

Briefly describe the different types of joints in robotics.

Answer 6(i)

Answer

Prismatic (sliding/linear): movement along an axis without rotation

Revolute (hinge/rotational): rotation around a fixed axis

Spherical (ball-and-socket): multi-directional rotation (pitch, yaw, roll)

Question 6(ii)

What is data theft? Give any two causes of data theft.

Answer 6(ii)

Answer

Data theft means acquiring someone's personal data (financial credentials, IP, confidential documents).
Causes: Hacking; Malware.

Question 6(iii)

Write a Python program to:
Create a list of 10 numbers.
Print elements at 2nd and 4th index.
Sort the list in ascending order.
Ask user for a number and search if it exists.

Answer 6(iii)

Answer

Create list, print `lst[2]` and `lst[4]`, call `lst.sort()`, then use `if num in lst`.

Question 7(i)

Explain the role of controllers in a robotic system. How do they work together with sensors and actuators?

Answer 7(i)

Answer

Controller manages overall functioning, processes input and gives instructions.
Sensors collect data $\rightarrow$ Controller processes and decides $\rightarrow$ Actuators carry out the action.

Question 7(ii)

Briefly explain supervised learning.

Answer 7(ii)

Answer

Model trained on labelled dataset (inputs + known outputs). Data split into training and test sets. Model learns from known results and predicts for unseen inputs.

Question 7(iii)

Write a Python program that performs the following tasks on a tuple:
Create tuple (12, 27, 30, 47, 55, 49), convert to list, remove 30, append [5, 25], insert 11 at index 3, print list, convert back to tuple.

Answer 7(iii)

Answer

Use `list()`, `remove()`, `append()`, `insert()`, then `tuple()`.

Question 8(i)

What is Tinkercad? Give any two features of the Tinkercad workspace.

Answer 8(i)

Answer

Tinkercad is a web-based design and modelling tool for 3D design, electronics and coding.
Features: drag-and-drop interface for 3D models; simulation of electronic circuits and robotic components.

Question 8(ii)

Give any three different ways to represent data to gather meaningful information.

Answer 8(ii)

Answer

Bar Graph: compare quantities across categories

Pie Chart: show parts of a whole / percentage distribution

Line Graph: show changes or trends over time

Question 8(iii)

Write a Python program that performs the following operations on a string:
Define “Machine Learning”, convert to lowercase, replace “Machine” with “Deep”, check if ends with “Learning”, count occurrences of letter “n”.

Answer 8(iii)

Answer

Use `lower()`, `replace()`, `endswith()`, `count('n')`.

Thank You!